



Faculty of Computing
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SE301.3 : SW Process Management

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Outline for Next two weeks

- **Introduction to Project Management**
 - Defining Project Success, Project Management
 - The Project Manager–Line Manager Interface,
 - Defining the Project Manager's Role,
 - The Project Manager as the Planning Agent,
 - Project Champions



References

- Project Management: A Systems Approach to Planning, Scheduling, and Controlling, Harold R. Kerzner, John Wiley and Sons, 2009, 10th Edition.
- A Guide to the Project Management Body of Knowledge (PMBOK® Guide), Project Management Institute, 2013, 5th Edition.
- Other Research papers – on class delivery



What is called Project

- A project is “a temporary endeavour undertaken to create a unique product, service, or result” (PMBOK® Guide, Fifth Edition, 2012)
- Operations is work done to sustain the business Projects end when their objectives have been reached or the project has been terminated
- Projects can be large or small and take a short or long time to complete
Information Technology Project Management, Seventh Edition



Project

- Project are
 - Performed by people.
 - Constrained by limited resources.
 - Planned, executed, and controlled.



Examples of Projects

- A team of students creates a smartphone application and sells it online
- A company develops a driverless car
- A small software development team adds a new feature to an internal software application for the finance department
- A college upgrades its technology infrastructure to provide wireless Internet access across the whole campus



Software Project Management

- Concerned with activities involved in ensuring that software is delivered on time and on schedule and in accordance with the requirements of the organisations developing and procuring the software.
- Project management is needed because software development is always subject to budget and schedule constraints that are set by the organisation developing the software.



Success Criteria

- Deliver the software to the customer at the agreed time.
- Keep overall costs within budget.
- Deliver software that meets the customer's expectations.
- At the desired performance/ technology level
- Maintain a coherent and well-functioning development team.

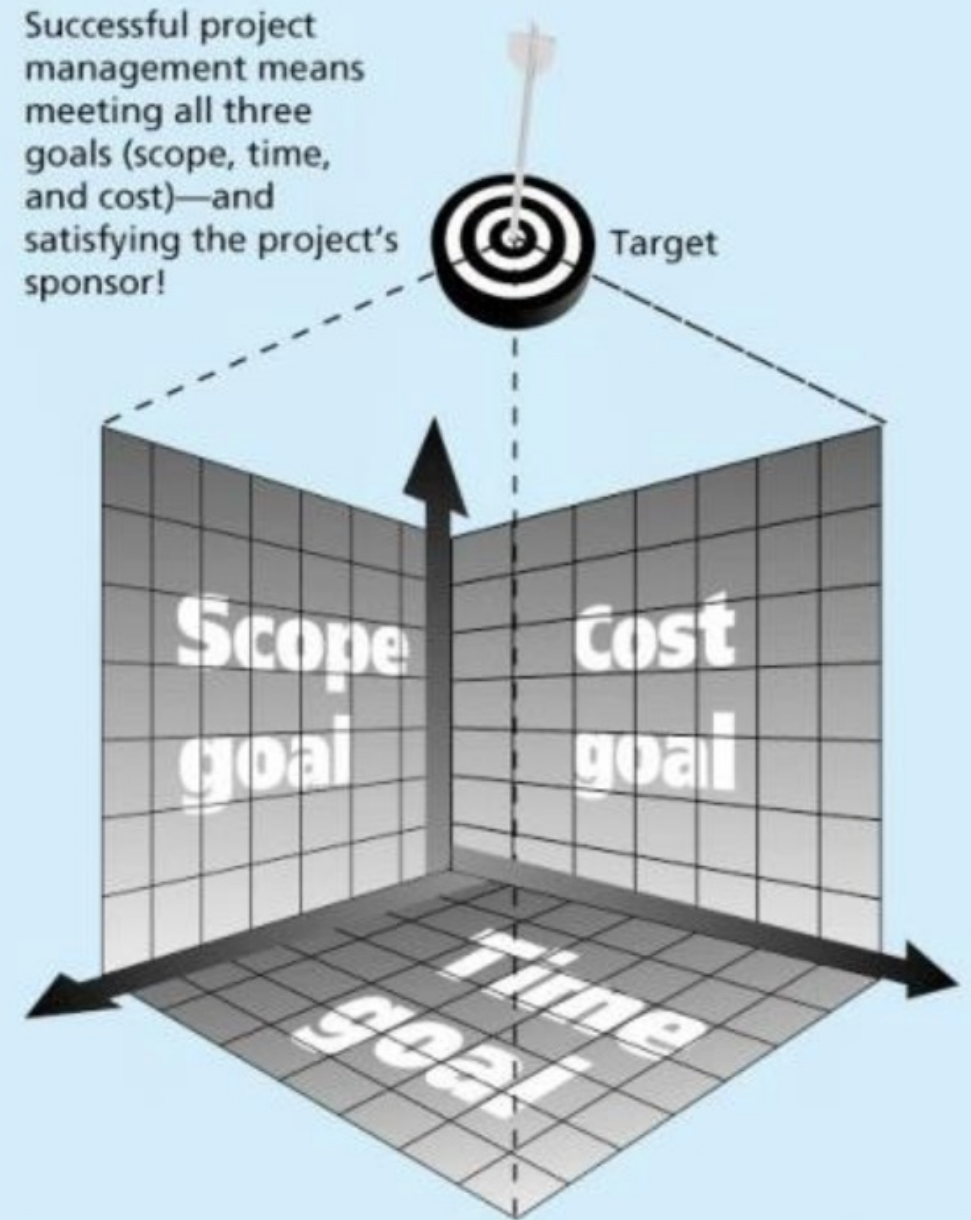


Software Management Distinctions

- The product is intangible.
 - Software cannot be seen or touched. Software project managers cannot see progress by simply looking at the artefact that is being constructed.
- Many software projects are 'one-off' projects.
 - Large software projects are usually different in some ways from previous projects. Even managers who have lots of previous experience may find it difficult to anticipate problems.
- Software processes are variable and organization specific.
 - We still cannot reliably predict when a particular software process is likely to lead to development problems.



The Tripple Constraint of a Project Management





Factors influencing Project Management

- Company size
- Software customers
- Software size
- Software type
- Organizational culture
- Software development processes
- These factors mean that project managers in different organizations may work in quite different ways.



Understanding the Project Management

- A project can be considered to be any series of activities and tasks that :
 - Have a specific objectives to be completed within certain specifications
 - Have defined start and end dates
 - Have funding limits
 - Consume human and nonhuman resources(money, people, equipment)
 - Multifunctional (like cut across several functional lines)



Potential benefits are

- Identification of functional responsibilities to ensure that all activities are accounted for, regardless of personnel turnover
- Minimizing the need for continuous reporting
- Identification of time limits for scheduling
- Identification of a methodology for trade-off analysis
- Measurement of accomplishment against plans
- Early identification of problems so that corrective action may follow
- Improved estimating capability for future planning
- Knowing when objectives cannot be met or will be exceeded



Overcoming Obstacles

- Project complexity
- Customer's special requirements and scope changes
- Organizational restructuring
- Project risks (Risk Management)
- Changes in technology
- Forward planning and pricing



Universal Management Activities

- *Project planning*
 - Project managers are responsible for planning, estimating and scheduling project development and assigning people to tasks.
- *Risk management*
 - Project managers assess the risks that may affect a project, monitor these risks and take action when problems arise.
- *People management*
 - Project managers have to choose people for their team and establish ways of working that leads to effective team performance.



Universal Management Activities

- *Reporting*
 - Project managers are usually responsible for reporting on the progress of a project to customers and to the managers of the company developing the software.
- *Proposal writing*
 - The first stage in a software project may involve writing a proposal to win a contract to carry out an item of work. The proposal describes the objectives of the project and how it will be carried out.



PBOK based Project Management Activities

- **Project Initiation**
 - Selection of the best project given resource limits
 - Recognizing the benefits of the project
 - Preparation of the documents to sanction the project
 - Assigning of the project manager
- **Project Planning**
 - Definition of the work requirements
 - Definition of the quality and quantity of work
 - Definition of the resources needed
 - Scheduling the activities
 - Evaluation of the various risks



PBOK based Project Management Activities

- **Project Execution**
 - Negotiating for the project team members
 - Directing and managing the work
 - Working with the team members to help them improve
- **Project Monitoring and Control**
 - Tracking progress
 - Comparing actual outcome to predicted outcome
 - Analyzing variances and impacts
 - Making adjustments

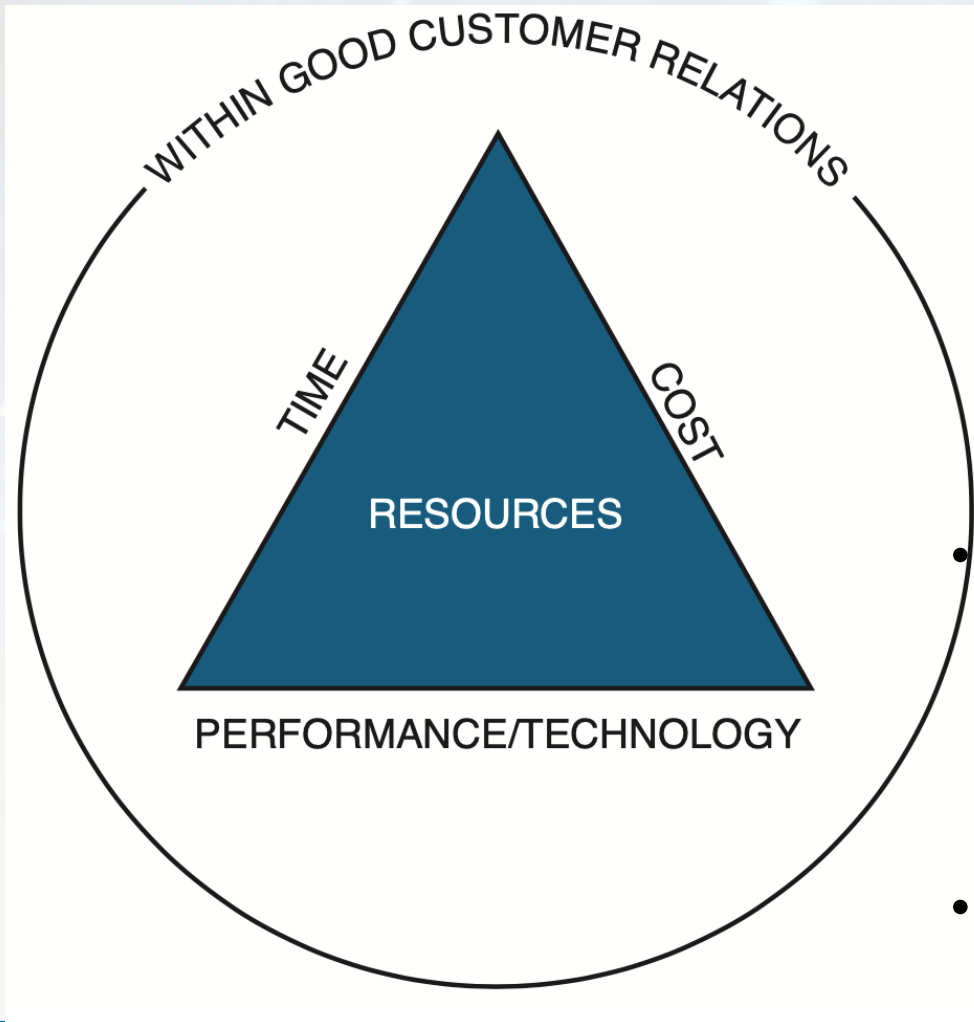


PBOK based Project Management Activities

- **Project Closure**
 - Verifying that all of the work has been accomplished
 - Contractual closure of the contract
 - Financial closure of the charge numbers
 - Administrative closure of the paper work



Overview of Project Management



- Figure Highlights that PM is designed to manage or control company resources on a given activity within time within cost and with in performance.
- Time, Cost and Performance are the constraints on the project



How PM can be defined now

- PM is planning, organising, directing, and controlling of company resources for a relatively short-term objective that has been established to complete specific goals and objectives.



Project Deliverables

- Projects exist to produce deliverables. The Person ultimately assigned as the project manager may very well be assigned based upon the size, nature, and scope of the deliverables.
- Deliverables are outputs, or the end result of either the completion of the project or the end of a lifecycle phase of the project. Importantly they are measurable, tangible outputs and can take such form as



Project Deliverables

- Hardware deliverables : Hardware items, a prototype, a piece of equipment
- Software Deliverables : these items are paper products, reports, studies, handouts, or documents ,
- Interim deliverables - can be either software or hardware deliverables and progressively evolve as the project proceeds.
interim reports leading to the final report



Thank you.

Profile



<https://www.sab.ac.lk/staff-directory/216>



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